SAT Math Mastery · Extra Practice

Thin Skills: Statistics, Probability, Functions and Similarity

Time about 2 hours: Part A in class, Part B independently.

Topics evaluating claims · margin of error · probability · two-variable data · composition and transformations of functions · sums and products of roots · similar triangles and area ratios

Name / Date _____ _____

PART A · IN CLASS — 25 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · NO CALCULATOR NEEDED

A researcher recruited 120 volunteers who responded to an online advertisement. She randomly assigned 60 of them to study with method X and 60 to study with method Y for a month. The method X group had a higher mean score on a final test, and the difference was larger than would be expected from random assignment alone. Which conclusion is best supported by the study?

(A) The study provides no evidence about either method.

(B) Method X caused higher scores for volunteers like those in the study.

(C) Method X is associated with higher scores, but no causal conclusion is possible.

(D) Method X causes higher scores for all students.

2. Hard · No calculator needed

A random sample of 500 adults was selected from a large city, and each adult reported how many hours per week they exercise and how many hours they sleep each night. The adults who exercise more tend to sleep more. Which conclusion is best supported by this study?

(A) Exercising more causes adults in the city to sleep more.

(B) Sleeping more causes adults in the city to exercise more.

(C) Among the 500 adults only, exercise and sleep are associated; nothing can be said about the city.

(D) Among adults in the city, exercise and sleep are positively associated, but exercise has not been shown to cause more sleep.

3. Hard · No calculator needed

In a poll of a random sample of voters, 51% said they support a proposed tax. The margin of error was 2.5% at the 95% confidence level. Based on the poll, which statement is best supported?

(A) A majority of all voters support the tax.

(B) It is plausible that a majority support the tax, but less than half supporting it is also plausible.

(C) Fewer than half of all voters support the tax.

(D) Exactly 51% of all voters support the tax.

4. Hard · No calculator needed

A random sample of 500 residents of Town P gave an estimate of 30% for the percent who favor a new park, with a margin of error of 4%. A random sample of 500 residents of Town Q gave an estimate of 38%, also with a margin of error of 4%. Which statement is best supported?

(A) It is plausible that the percent in favor is the same in both towns.

(B) The percent in favor in Town P is exactly 4 points less than that in Town Q.

(C) Town Q definitely has a greater percent in favor than Town P.

(D) It is not possible for the percents to be equal.

5. Hard · No calculator needed

A school has 1,200 seniors and 2,800 students in other grades. A random sample of 300 seniors was surveyed, and an estimated 62% plan to attend college, with a margin of error of 5%. Which statement is best supported by the survey?

(A) Exactly 62% of the seniors at the school plan to attend college.

(B) Between 57% and 67% of the 300 surveyed seniors plan to attend college.

(C) Between 57% and 67% of all students at the school plan to attend college.

(D) It is plausible that between 57% and 67% of the seniors at the school plan to attend college.

6. Hard · Desmos

Events A and B are independent. The probability of A is 0.4, and the probability that at least one of A and B occurs is 0.7. What is the probability of B?

7. Hard · Desmos

Two fair six-sided dice are rolled. Given that at least one of the dice shows an even number, what is the probability that the sum of the two dice is 8?

8. Hard · Desmos

In a town, 30% of the residents are over 60 years old. Of the residents over 60, 80% vote in elections; of the residents who are not over 60, 50% vote. A voter is chosen at random from all residents who vote. What is the probability that the voter is over 60?

9. Hard · Desmos

The table shows values of x and y for a data set.

x	0	1	2	3	4
y	8	12	18	27	40.5

Which equation fits all five pairs?

(A) $y = 8(1.5)^x$

(B) $y = 4x + 8$

(C) $y = 2x^2 + 2x + 8$

(D) $y = 8(1.2)^x$

10. HARD · DESMOS

Ten data points lie exactly on the line $y = 2x + 1$, at $x = 1, 2, \ldots, 10$. A new data point $(11, 40)$, far above that line, is added. How does the least-squares line for all eleven points compare with the line $y = 2x + 1$?

(A) Smaller slope and greater y-intercept

(B) The same slope and the same y-intercept

(C) Greater slope and smaller y-intercept

(D) Greater slope and greater y-intercept

11. HARD · DESMOS

The residuals for the first four of five data points about a least-squares line are -2, 3, 1 and -4. What is the residual for the fifth point?

12. HARD · DESMOS

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 5$. The function g is defined by $g(x) = f(2x)$. What are the x-intercepts of the graph of $y = g(x)$?

(A) $x = -2$ and $x = \dfrac{5}{2}$

(B) $x = -4$ and $x = 10$

(C) $x = -1$ and $x = \dfrac{5}{2}$

(D) $x = 0$ and $x = 3$

13. Hard · Desmos

The functions f and g are defined for the values in the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	4	1	2	3

What is the value of $g\left(f^{-1}(g(3))\right)$?

14. Hard · Desmos

The function f is odd, which means $f(-x) = -f(x)$ for every x. Which of the following functions must be even, meaning $h(-x) = h(x)$ for every x?

(A) $[f(x)]^2$

(B) $f(f(x))$

(C) $f(x) + 1$

(D) $f(x) + x$

15. Hard · Desmos

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 - 5$. Which describes all real values of x for which $f(g(x))$ is defined?

(A) $x \geq 1$

(B) $-\sqrt{6} \leq x \leq \sqrt{6}$

(C) all real numbers

(D) $x \leq -\sqrt{6}$ or $x \geq \sqrt{6}$

16. HARD · DESMOS

The graph of $y = f(x)$ passes through the point $(4, -2)$. Which point must lie on the graph of $y = 3f(x - 1) + 5$?

(A) $(5, 11)$

(B) $(5, -1)$

(C) $(3, -1)$

(D) $(3, 11)$

17. HARD · DESMOS

The equation $2x^2 - 7x + k = 0$, where k is a constant, has two real solutions r and s with $r^2 + s^2 = 9$. What is the value of k?

18. HARD · DESMOS

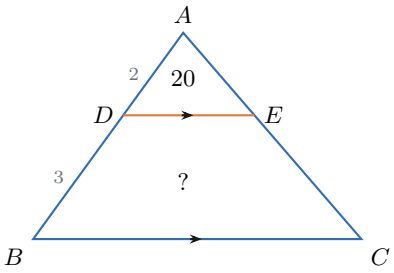
What is the sum of all real solutions of $2^{2x} - 6 \cdot 2^x + 8 = 0$?

19. HARD · DESMOS

Triangle ABC is similar to triangle DEF. The area of triangle ABC is 36 and the area of triangle DEF is 100. The perimeter of triangle ABC is 24. What is the perimeter of triangle DEF?

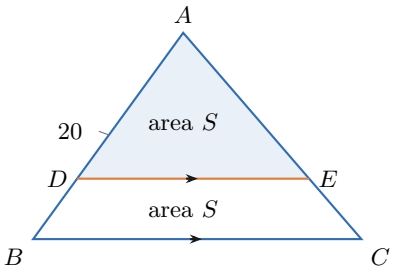
20. Hard · Desmos

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD : DB = 2 : 3$ and the area of triangle ADE is 20, what is the area of quadrilateral $DBCE$?



21. Hard · Desmos

In triangle ABC, point D is on $\overline{AB}$ and point E is on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. The segment $\overline{DE}$ divides the triangle into two regions of equal area. If $AB = 20$, what is the length of $\overline{AD}$, to the nearest tenth?



22. Hard · Desmos

A square pyramid has a base with area 144. A plane parallel to the base cuts the pyramid at a point one-third of the way from the apex to the base, forming a smaller square cross-section. What is the area of the cross-section?

23. Hard · Desmos

The graph of $y = f(x)$ has x-intercepts at $x = 1$ and $x = 6$. The function g is defined by $g(x) = f(3 - x)$. What is the greatest x-intercept of the graph of $y = g(x)$?

24. Hard · Desmos

The solutions of $x^2 - 5x + 2 = 0$ are r and s. What is the value of $\frac{r}{s} + \frac{s}{r}$?

25. Hard · Desmos

The polynomial $p(x) = x^3 - 7x^2 + 14x - 8$ has three real zeros. What is the sum of the squares of the zeros?

PART B · SELF-ASSESSMENT — 20 PROBLEMS

26. HARD · DESMOS

A random sample of students at a university reported a mean of 3.2 hours of study per day, with a margin of error of 0.4 hours. The university has 18,000 students. Based on the sample, what is the greatest total number of study hours per day that is plausible for the whole student body?

27. HARD · DESMOS

To estimate the number of fish in a lake, a biologist caught and tagged 150 fish and released them. A week later she caught a random sample of 60 fish from the lake, and 9 of them were tagged. Assuming the sample is representative and the tagged fish mixed evenly, what is the best estimate of the number of fish in the lake?

28. HARD · DESMOS

A company's customers were asked whether they renewed their contract. The results are shown in the table.

	Renewed	Did not renew
Plan A	84	36
Plan B	45	35

If one of the customers who did not renew is chosen at random, what is the probability that the customer was on Plan B?

29. Hard · Desmos

A bag holds 5 red marbles and 3 blue marbles. Two marbles are drawn at random, one after the other, without replacement. What is the probability that the two marbles are the same color?

(A) $\frac{15}{28}$ (B) $\frac{13}{28}$

(C) $\frac{5}{14}$ (D) $\frac{17}{32}$

30. Hard · Desmos

A line of best fit for a data set is $\hat{y} = 3.2x + 14$. For the data point with $x = 10$, the actual value is $y = 51$. What is the residual for this point (actual value minus predicted value)?

31. Hard · Desmos

A least-squares line for a data set is $\hat{y} = 1.5x + 4$. The mean of the x-values is 10. What is the mean of the y-values?

32. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{2x+1}{x-3}$ for $x \neq 3$. What is the value of $f^{-1}(5)$?

33. HARD · DESMOS

What is the product of all real solutions of $x^4 - 5x^2 + 4 = 0$?

34. HARD · DESMOS

A right triangle has legs of length 6 and 8. A larger, similar right triangle has area 150. What is the length of the hypotenuse of the larger triangle?

35. HARD · DESMOS

The graph of the quadratic function f has vertex $(4, -8)$ and passes through the point $(0, 8)$. The function g is defined by $g(x) = f(2x)$. What is the value of $g(1)$?

36. Hard · Desmos

The function f is even, so $f(-x) = f(x)$ for every x. The function g is defined by $g(x) = f(x) + f(-x)$. If $f(3) = 7$ and $f(-5) = 2$, what is the value of $g(3) + g(5)$?

37. Medium · No calculator needed

A website for a cycling club posted a poll asking, "Should the city spend more on bike lanes?" Of the 1,850 people who chose to respond, 91% voted yes. Which is the main reason the poll result should not be used to estimate the opinion of all city residents?

(A) Respondents selected themselves and are likely to hold similar views.

(B) The sample size is too large.

(C) The question has too few possible answers.

(D) The poll was conducted online, and no online poll can be accurate.

38. Medium · No calculator needed

A random sample of 500 adults reported their coffee consumption. The adults who drink coffee daily had a lower rate of a certain heart condition than those who do not. Which change to the study would make it possible to conclude that drinking coffee reduces the rate of the condition?

(A) Increase the sample to 5,000 adults.

(B) Include only adults older than 40.

(C) Randomly assign the participants to drink coffee daily or not, then compare the rates.

(D) Ask each adult about more lifestyle habits.

39. Medium · Desmos

Two independent random samples were taken from the same population to estimate a proportion at the same confidence level. The first sample had 400 members and a margin of error of 6%. The second sample had 1,600 members. Which is most likely the margin of error for the second sample?

(A) 1.5%

(B) 6%

(C) 3%

(D) 12%

40. Medium · Desmos

Of 500 residents surveyed, 240 own a car, 180 own a bicycle, and 70 own both. If one of the surveyed residents is chosen at random, what is the probability that the resident owns neither a car nor a bicycle?

41. Medium · Desmos

In a trial, 90 of 120 patients given Drug X improved, and 60 of 100 patients given Drug Y improved. How many times as likely is a patient given Drug X to improve as a patient given Drug Y?

42. Medium · Desmos

A teacher modeled the final score y of students in a course with the line of best fit $\hat{y} = 82 - 1.6x$, where x is the number of days the student was absent. Which statement is the best interpretation of the number 82 in this model?

(A) A student absent for 82 days is predicted to score 0.

(B) The predicted score decreases by 82 for each day absent.

(C) The mean score of all students in the class is 82.

(D) The predicted score of a student with no absences is 82.

43. Medium · Desmos

A model $y = 0.85x + 12$ was fitted to data in which x ranged from 20 to 80. A student uses the model to predict y when $x = 150$. Which statement about this prediction is best?

(A) It is reliable, because the slope is positive.

(B) It is less reliable, because 150 is outside the range of the data.

(C) It is impossible, because y cannot be computed for $x > 80$.

(D) It is reliable, because the model is linear.

44. Medium · Desmos

The function f is defined by $f(x) = (x - 1)(x - 5)$. The graph of g is the reflection of the graph of f across the y-axis, shifted up 4 units. What is the value of $g(2)$?

45. Medium · Desmos

Two similar triangles have areas 18 and 50. The altitude to the longest side of the larger triangle is 15. What is the corresponding altitude of the smaller triangle?

(A) 10.8

(B) 5.4

(C) 12.5

(D) 9